Beyond a Single Structure: Exploring Multistate Protein Conformational Landscapes with AI Structure Prediction

T.L. Benavides[1], G.T. Montelione[2] · montelionelab@rpi.edu

[1] Biological Sciences, Rensselaer Polytechnic Institute [2] Department of Chemistry and Chemical Biology, Rensselaer Polytechnic Institute

Abstract

Proteins function through conformational ensembles. To assess the ability of current foundational structure inference models to predict multiple protein states, we generated 22,000 monomer models (11,000 per target; 1,000 per sampling method per target) for E. coli Adk (PDB IDs 4AKE/1AKE)[1,2] and Mad2 mitotic spindle assembly checkpoint protein (PDB IDs 1DUJ/1S2H)[3,4] using 11 different Artificial Intelligence Sampling (AIS) methods. Four AIS methods involve applying a stochastic 40% MSA column masking to foundational structure inference models, adapted from Kalakoti and Wallner (Fig. 1a).[5] Shared Cα pair distances from the AIS models were used as features for clustering with UMAP, PCA, t-SNE and Bonsai. These AIS procedures produced models with best TM-scores (US-align) of 0.97/0.99 against open/closed adenylate kinase references (4AKE/1AKE) and 0.80/0.83 against open/closed Mad2 references (1DUJ/1S2H). Models were used as templates with Boltz2 in no-MSA mode to retrieve reported pTM scores, similar to the AFRank methodology proposed by Roney and Ovchinnikov (Fig. 1b).[6] Each target was analyzed on four A100 GPUs in approximately 24 hours. This AIS protocol is continued on a set of 500+ protein targets with experimental evidence of multiple states. All AIS models and Boltz2 template-only refolds for the two targets presented here are deposited on Zenodo:
Adenylate kinase: 10.5281/zenodo.23601146
Mad2: 10.5281/zenodo.23601144
Code Accessibility: https://github.com/benavt/SCHERAGA_POSTER
A virtual poster accompanies this physical poster, accessible by QR code.

Background

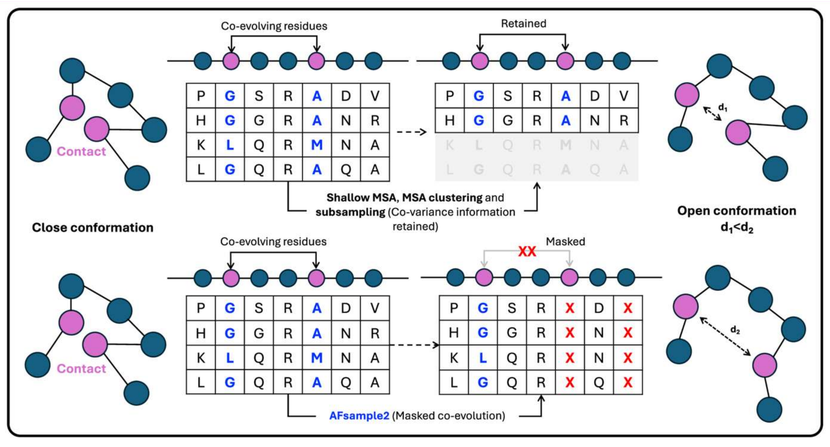


Figure 1a. MSA subsampling and masking. Kalakoti & Wallner (2025), Fig. 1a[5], CC BY 4.0.

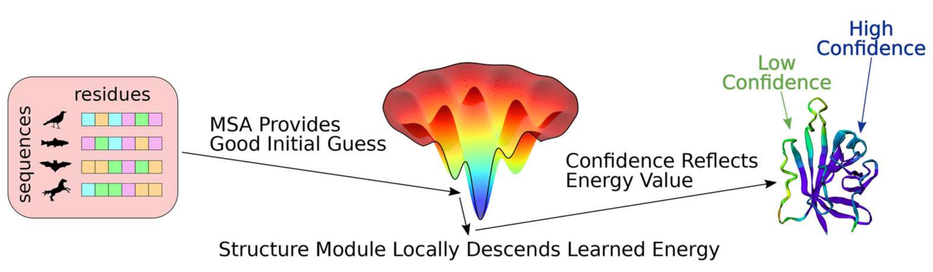


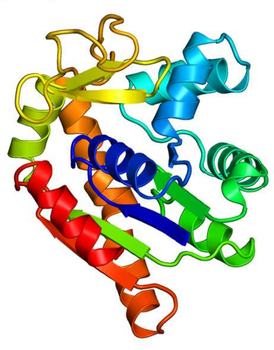
Figure 1b. Learned structure landscape. From Roney & Ovchinnikov (2022), Fig. 1[6]. CC BY 4.0.

A / E. coli adenylate kinase

S1 · AlphaFlow-MD
1AKE_alphaflow-md_0703.pdb



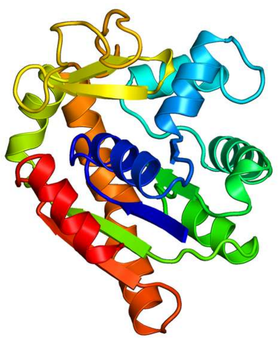
S2 · AlphaFlow-PDB
1AKE_alphaflow-pdb_0327.pdb



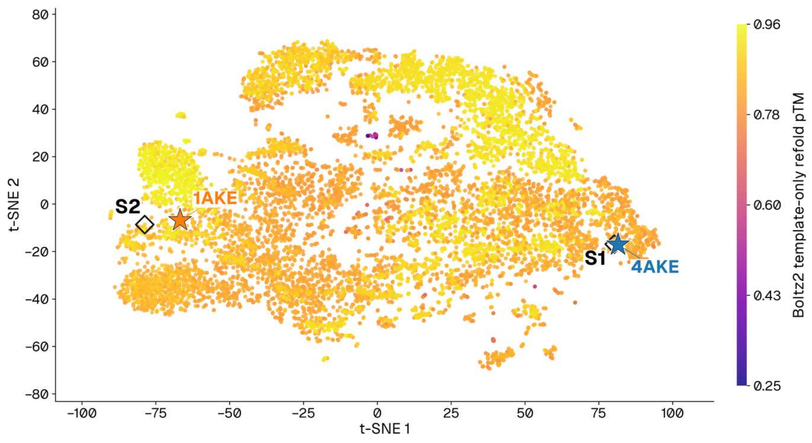
4AKE open



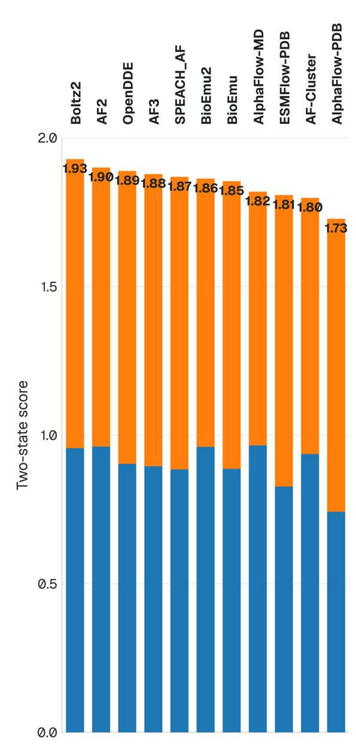
1AKE closed



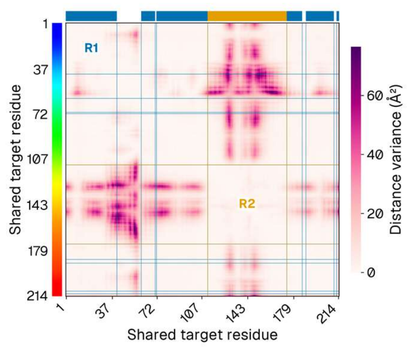
A1. AI sampling produces open and closed states. TM/Boltz2 refold pTM: S1 (4AKE[1]) 0.97/0.82; S2 (1AKE[2]) 0.99/0.88.



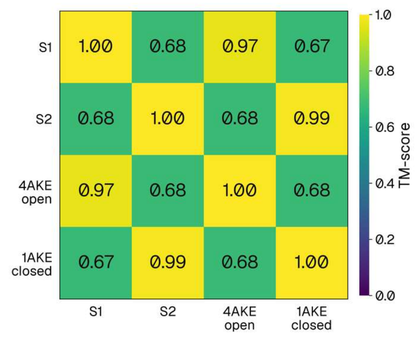
A2. t-SNE projection. 11,000 models; shared pairwise Cα distances, colored by Boltz2 template-only refold pTM. ◇ S1/S2 ★ 4AKE open[1] ★ 1AKE closed[2]



A5. Two-state score. Best TM(■ 4AKE open[1]) + best TM(■ 1AKE closed[2]), over 1,000 models per method. Methods: **BioEmu**, **BioEmu2**, **SPEACH_AF**, **AF-Cluster**, **AlphaFlow-PDB**, **AlphaFlow-MD**, **ESMFlow-PDB**, and 40% stochastic column masking applied to **AF2**, **AF3**, **OpenDDE**, and **Boltz2**.



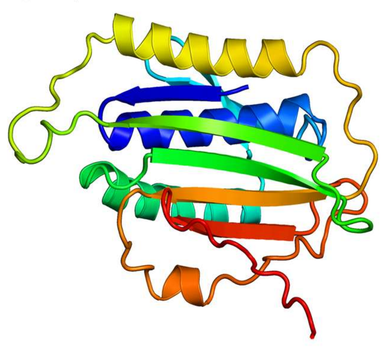
A3. Cα distance variance. R1: core anchor used for structural alignment. R2: A112–173 moving comparison region.



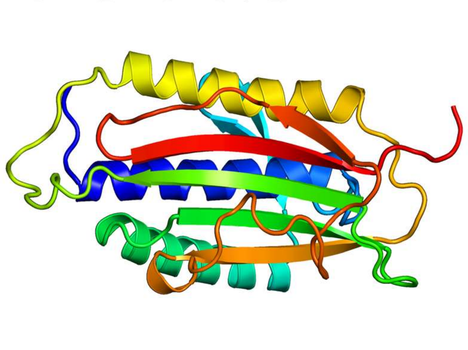
A4. TM-score matrix. S1/S2 and PDB controls. 214 shared Cα. Best AIS TM: 4AKE open[1] 0.97; 1AKE closed[2] 0.99.

B / Mad2 mitotic spindle checkpoint protein

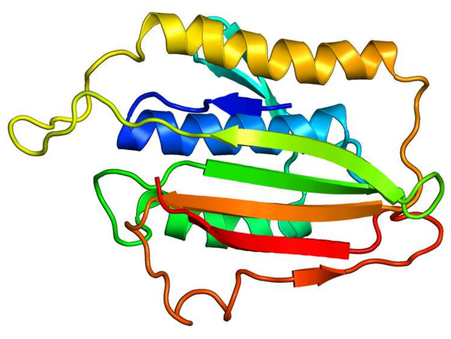
S1 · AF-Cluster
MAD2_afcluster_0675.pdb



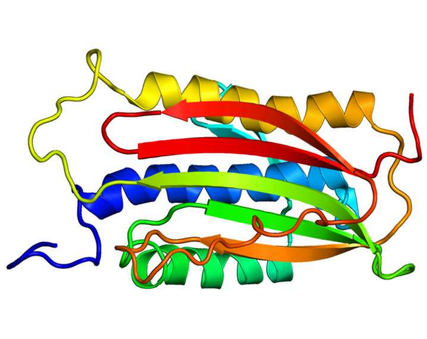
S2 · Boltz2
1KLQ_MAD2_ALONE__Boltz2_Exact_40pct__0852.cif



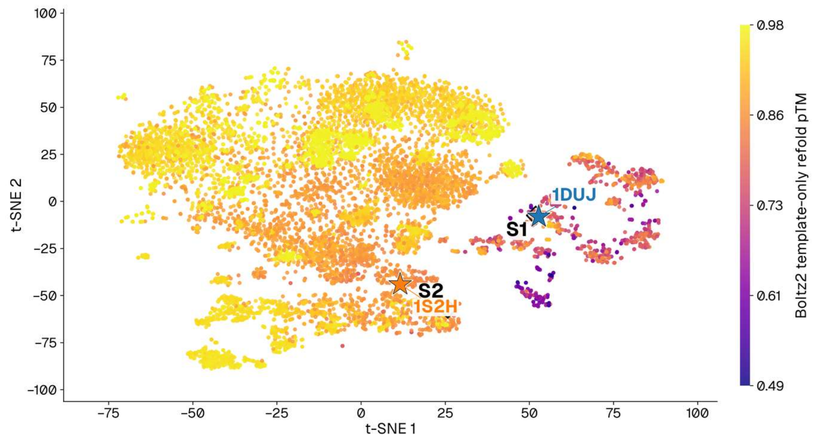
1DUJ open



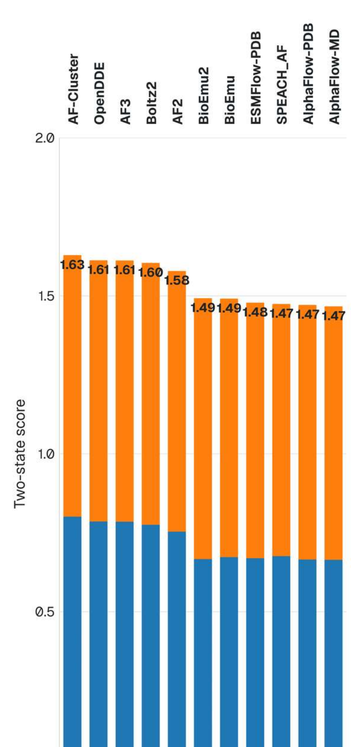
1S2H closed



B1. AI sampling produces open and closed states. TM/Boltz2 refold pTM: S1 (1DUJ[3]) 0.80/0.89; S2 (1S2H[4]) 0.83/0.94.

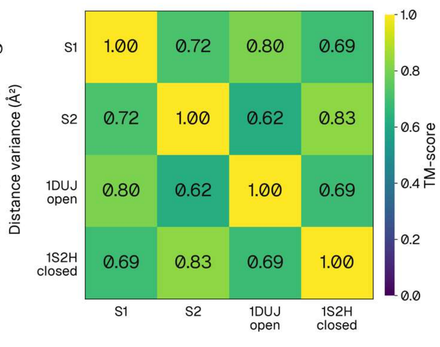


B2. t-SNE projection. 11,000 models; shared pairwise Cα distances, colored by Boltz2 template-only refold pTM. ◇ S1/S2 ★ 1DUJ open[3] ★ 1S2H closed[4]



B5. Two-state score. Best TM(■ 1DUJ open[3]) + best TM(■ 1S2H closed[4]), over 1,000 models per method.

B3. Cα distance variance. R1: core anchor used for structural alignment. R2: shared C-terminal residues A151–187.



B4. TM-score matrix. S1/S2 and PDB controls. 175 shared Cα. Best AIS TM: 1DUJ open[3] 0.80; 1S2H closed[4] 0.83.

References

Additional references can be found in the virtual poster.

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VIRTUAL POSTER

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